

EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge MA

Candidate for B.S. in Mechanical Engineering with Concentration in Computer Science and Minor in German

May 2027

- GPA: 5.0/5.0
- Varsity Basketball Starter, Offensive Defensive Player of the Week winner, Defensive Player of the Week winner
- Relevant Coursework: Robotics, Dynamics and Control, Machine Learning, Design and Manufacturing, Thermo- and Fluid Dynamics, Mechanics and Materials, Fundamentals of Programming, Numerical Computation

PROFESSIONAL EXPERIENCE

GE Vernova

Greenville, SC

Modeling and Simulation, Controls Intern

May 2026 – August 2026

- Customized gas turbine simulations for customer-specific use cases, including analyzing how different control inputs affect relevant outputs, utilizing Model-Based Control, and integrating first-principles based modeling techniques
- Developed controls setups, modeling software, and simulation configurations for real-world gas turbine sites
- Worked in a team-based environment, collaborating with full-time engineers to across disciplines

MIT DeCoDe Lab

Cambridge, MA

Undergraduate Researcher

August 2025 – May 2026

- Built pipeline to automatically extract programmatic sequences from CAD data to facilitate Machine Learning endeavors
- Developed automatic process to create dataset of videos of GUI actions to construct CAD parts in order to enable training of an agentic model with the ability to manipulate mouse and keyboard to create complex CAD geometries

Mecado (VC-Backed AI Startup)

Cambridge, MA

Machine Learning Engineer

June 2025 – February 2026

- Led the initiative to adapt and train a Graph Neural Network-based architecture for semantic labeling of faces on 3D CAD parts and assemblies, including integrating a novel inference mechanism into CAD validation software
- Benchmarked SOTA 3D learning models, adjusting and editing training parameters and architecture methodologies
- Utilized Reinforcement Learning and LLM fine-tuning to develop a text-to-CAD model integrated into Siemens NX

DESY CMS Group

Hamburg, Germany

Research Assistant

June 2024 – August 2024

- Investigated ALPs (Axion-Like Particles) and their potential signatures at the FCC-ee (Future Circular Collider—electron-positron) using Monte-Carlo simulations, ROOT graphing tools, and statistical and physical analyses
- Published 30-page paper summarizing new research in the field of BSM (Beyond the Standard Model) particle physics
- Presented a final research presentation to the global FCC LLP group; developed and updated CERN FCCAnalyses repository

MIT Atomistic Simulation and Energy (ASE) Research Group

Cambridge MA

Undergraduate Researcher

February 2024 – May 2024

- Characterized the thermal and mechanical properties of graphite for applicability in heat-based energy storage methods
- Performed mechanical testing before and after heating up graphite samples to discover results of extreme heat exposure

LEADERSHIP

MIT Teaching Assistant

Cambridge MA

Physics: Mechanics; Physics: Electricity and Magnetism; Electronics for Mechanical Systems

September 2024 – Present

- Teaching problem-solving strategies and helping 60+ students work through group problems and PSETs to develop a fundamental understanding of physics, from Newton's Laws to Maxwell's Equations
- Mentoring 30+ students with the building and debugging of electronic circuits across various projects

SKILLS & PROJECTS

Languages: German (advanced), Spanish (intermediate)

Programming: Python (+ PyTorch, Scikit, Numpy, Pandas, ROOT), C/C++, Java, JavaScript, Fortran, MATLAB, LaTeX, Git/Github

Software: CAD (SolidWorks, NX, Onshape), Ansys (Mechanical, Discovery, Workbench), MATLAB/Simulink, OpenModelica

Portfolio: merlin-gogolin.github.io/merlin-gogolin

Projects and Publications:

- Acted as team lead for MIT robotics class, utilizing kinematics, controls principles, and CV to program a UR5 robot arm and design and fabricate a custom end-effector to complete complex household tasks in a quick and consistent manner
- Performed experiment on how finger placement affects guitar sound quality, culminating in academic paper and poster
- Held the highest score by end of semester in robotics competition for historic MIT Design and Manufacturing class (2.007)
- Hold all-time record in line-following robot competition in Electronics for Mechanical Systems class
- MIT Sandbox funded full-stack app to intelligently recommend news articles to users, utilizing embedding techniques
- Coding projects: created an IDE from the ground up for Scheme coding language; coded an autocorrector to intelligently guess similar words and phrases; built ML-based fantasy football mock draft simulator and player score predictor
- 30-page physics publication: "Simulating and Investigating Long-Lived Axion-Like Particles at the Proposed FCC-ee": indico.desy.de/event/46018/contributions/173511/attachments/91780/124063/Merlin_Summer_Report_ALPs_at_the_FCCee